

IGITABO CYUZUYE: OMNISCIENCE FRAMEWORK

Ubushobozi, Ibyakora, n'Uburyo bwo Gutwikira no Kurinda

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Contents

INTANGIRIRO	2
1 UKO SISITEMU IKORA	3
1.1 Ibyiringiro kuri Sisitemu	3
1.2 Iyakora Sisitemu	3
2 UBUSHOKOZI N'AMAKURU	4
2.1 Ubushakashatsi ku makuru	4
2.2 Amakuru yifashishwa	4
3 IBIKORESHO BYAKORESHEJWE	5
3.1 Ibyiringiro	5
4 UBUSHOBOZI MU GUTWIKIRA	6
4.1 Ukwifashisha mu buryo budasanzwe	6
4.2 Uburyo bwo Kurinda	6
5 IBIKURIKIRAHU N'IBYIFASHISHWA MU RWANDA	7
ICYUZUYO	8
GUSOZA	9

INTANGIRIRO

Muri iki gitabo, tugiye gutanga ubushobozi bwuzuye bwigerageje neza kurushaho kubona ikintu cyemewe cyifashishwa mu bumenyi bw'ikoranabuhanga n'ubucuti bw'ikoranabuhanga mu Rwanda n'ahandi hose. Twifashishije ikoranabuhanga rishya ryitwa "Omniscience Framework" (urubuga rwuzuye rw'ubumenyi), turagaragaza uburyo bwo gukoresha ubu bushobozi mu buryo bukenewe mu kwifashisha mu macukurushaka n'ibindi bifashishwa mu bumenyi.

Iki kibano kigaragaza ibitekerezo byuzuye kuri iki gikorwa cyanze mu Kinyarwanda. Twifashishije imvo myinshi yagaragaye mu mategeko y'ikoranabuhanga, tugerageza kugaragaza uko iyi sisitemu ikora, n'uko abantu bashobora gukoresha ubu bushobozi mu buryo bukenewe.

Chapter 1

UKO SISITEMU IKORA

1.1 Ibyiringiro kuri Sisitemu

Omniscience Framework ni porogaramu ifite ubushobozi budasanzwe mu bumenyi bw'ikoranabuhanga. Ifite impuguke zifashishwa mu gutegura, gukusanya, no gukurikirana amakuru anyuranye aherekeje ku rubuga rw'ikoranabuhanga. Sisitemu ikora mu buryo bu buryo bwo gutegura, gushushanya, no gukurikirana ibintu byose bikeneye mu bumenyi bw'ikoranabuhanga.

Sisitemu ikoresha uburyo bubiri bwigenga mu gukusanya amakuru. Ubwo bwoko bushaka amakuru afite ikibanza cyihariye mu kibano cy'ikoranabuhanga. Sisitemu ikoresha uburyo bugufi mu gushaka amakuru, ifashishije ibikoresho by'indimi zitandukanye zifashishwa mu gutunganya amakuru. Ibi bishoboza sisitemu kugira ubushobozi bwo gukora ibintu byinshi bitandukanye.

1.2 Ibyakora Sisitemu

Sisitemu ifite ubushobozi bwo gukora ibintu byinshi bitandukanye. Muri ibi, dukwifashisha ibikoresho bitandukanye byifashishwa mu gukurikirana n'ibibera kuri sisitemu. Sisitemu ikoresha uburyo bubiri bwigenga mu gukora ibintu:

- **Uburyo bwo gutegura:** Sisitemu ikoresha uburyo bubiri mu gutegura. Mbere, ifashishije ibikoresho byo mu ngufu yo gutegura. Kabiri, ikoresha uburyo bubiri bwigenga mu gutegura.
- **Uburyo bwo gukurikirana:** Sisitemu ifite ubushobozi bwo gukurikirana ibintu byose bikeneye. Ibi bishoboza kugira uruhushya rwuzuye rwifashishwa mu gukora ibintu bitandukanye.

Chapter 2

UBUSHOKOZI N'AMAKURU

2.1 Ubushakashatsi ku makuru

Sisitemu ifite ubushobozi budasanzwe mu gutunganya amakuru anyuranye. Mu buryo budasanzwe, sisitemu ikora ibikoresho bitandukanye byifashishwa mu gutunganya amakuru. Ibi birashobora gukoreshwa mu buryo budasanzwe mu bumenyi bw'ikoranabuhanga. Sisitemu ikoresha uburyo bubiri mu gutunganya amakuru:

- **Uburyo bwo gutunganya:** Sisitemu ikoresha uburyo bubiri bwigenga mu gutunganya amakuru. Ibi birashobora gukoreshwa mu buryo budasanzwe.
- **Uburyo bwo gukurikirana:** Sisitemu ifite ubushobozi bwo gukurikirana amakuru anyuranye. Ibi bishoboza kugira uruhushya rwuzuye rwifashishwa mu gutunganya amakuru.

2.2 Amakuru yifashishwa

Sisitemu ikusanya amakuru anyuranye. Muri ibi, dukwifashisha amakuru atandukanye:

- **Amakuru yerekeye abantu:** Sisitemu ikusanya amakuru yerekeye abantu.
- **Amakuru yerekeye ibikoresho:** Sisitemu ikusanya amakuru yerekeye ibikoresho.
- **Amakuru yerekeye ikoranabuhanga:** Sisitemu ikusanya amakuru yerekeye ikoranabuhanga.

Chapter 3

IBIKORESHO BYAKORESHEJWE

3.1 Ibyiringiro

Sisitemu ikoresha ibikoresho bitandukanye byifashishwa mu gukora ibintu bitandukanye.

- **Porogaramu yibanze:** Sisitemu ikoresha Python nk'ururimi rwibanze rwifashishwa mu gukora.
- **Impuguke zifashishwa (Libraries):**
 - **Scapy:** Ifashishwa mu gutegura no gukusanya amapaki y'ikoranabuhanga (network packets).
 - **Impacket:** Ifashishwa mu gukora ibintu bitandukanye byerekeye SMB na WMI.
 - **Paramiko:** Ifashishwa mu gukora ibintu byerekeye SSH.
 - **Nmap:** Ifashishwa mu gutegura no gushaka amakuru kuri sisitemu zitan-dukanye.

Chapter 4

UBUSHOBOZI MU GUTWIKIRA

4.1 Ukwifashisha mu buryo budasanzwe

Sisitemu ifite ubushobozi bwo gukoreshwa mu bumenyi bw'ikoranabuhanga. Abakozi ba ICT bashobora gukoresha iyi sisitemu mu gukora ibintu byinshi:

- Gushaka ibikoresho bifite ububugenge kuri network.
- Gukusanya amakuru yerekeye sisitemu zitandukanye.
- Gukurikirana ibikoresho byafashishwa mu gutegeza.

4.2 Uburyo bwo Kurinda

- **Kurinda sisitemu yawe:** Ugira uruhushya rwuzuye rwifashishwa mu kurinda sisitemu yawe binyuze mu gukurikirana network.
- **Kurinda abakoresha:** Kubona abakoresha bashakira gukoresha ububugenge no gukurikirana ibikoresho byabo.

Chapter 5

IBIKURIKIRAHU N'IBYIFASHISHWA MU RWANDA

Mu Rwanda, sisitemu ishobora gukoreshwa mu buryo budasanzwe mu mashuri, mu bigo by'ubucuruzi, ndetse no mu nzego z'ibanze kugira ngo hashyizweho uburyo bwo kurinda amakuru.

ICYUZUYO

Muri iki gitabo, twagaragaje ubushobozi budasanzwe bw'Omniscience Framework mu Kinyarwanda. Twifashishije imvo myinshi yagaragaye mu mategeko y'ikoranabuhanga, tugerageza kugaragaza uko iyi sisitemu ikora, n'uko abantu bashobora gukoresha ubu bushobozi mu buryo bukenwe.

GUSOZA

Iki gitabo cyanditswe mu Kinyarwanda kugira ngo gifashe abakozi ba ICT mu Rwanda gusobanukirwa no gukoresha ubu buryo mu mutekano w’amakuru.

*Reba no: [Uburyo bwo gukoresha iyi sisitemu], [Ibyifashishwa mu Kinyarwanda],
[Ibitekerezo byifashishwa]*